# Business Statistics

## Populations and samples

Datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

# Population and Samples
## Overview

- Selecting the data depends on the question.
- Population and Samples
- Descriptive and Inferential Statistics

# Start with the business question and business decision

# Business decisions
# The question drives the data needs

- There is concern that students in the United States are behind other countries in science, technology, engineering and math (STEM) skills
- Decision: Should additional time and money be invested to improve the curricula of analytics courses across the country?
- Question: What is the current performance of students in analytics courses?
  - How to measure performance?
  - Of which students?
  - In which courses?
  - Over what period of time?

# Population of interest
# Collecting the data to answer the question

- Decision: Should our schools' Business Analytics Department invest time and money to improve the current business statistics course?
- Question: What is the average GPA of students in our business statistics course over the past 5 years?
- Population of interest: our business statistics students over the past 5 years
- Required Data: GPA and demographics for members of the population
- Is it feasible and practical to collect data for the entire population? **Yes**
- Action: Collect data from the population of interest and answer the question.

# Population of interest
# Collecting the data to answer the question

- Decision: Should all universities across the United States invest additional time and money to improve current business statistics courses?
- Question: What is the average GPA of all business statistics students over the past 5 years?
- Population of interest: all US business statistics students over the past 5 years
- Required Data: GPA and demographics for members of the population
- Is it feasible and practical to collect data for the entire population? No
- Action: select a sample from the population of interest and then use that to draw conclusions about the population.

# Population and Samples

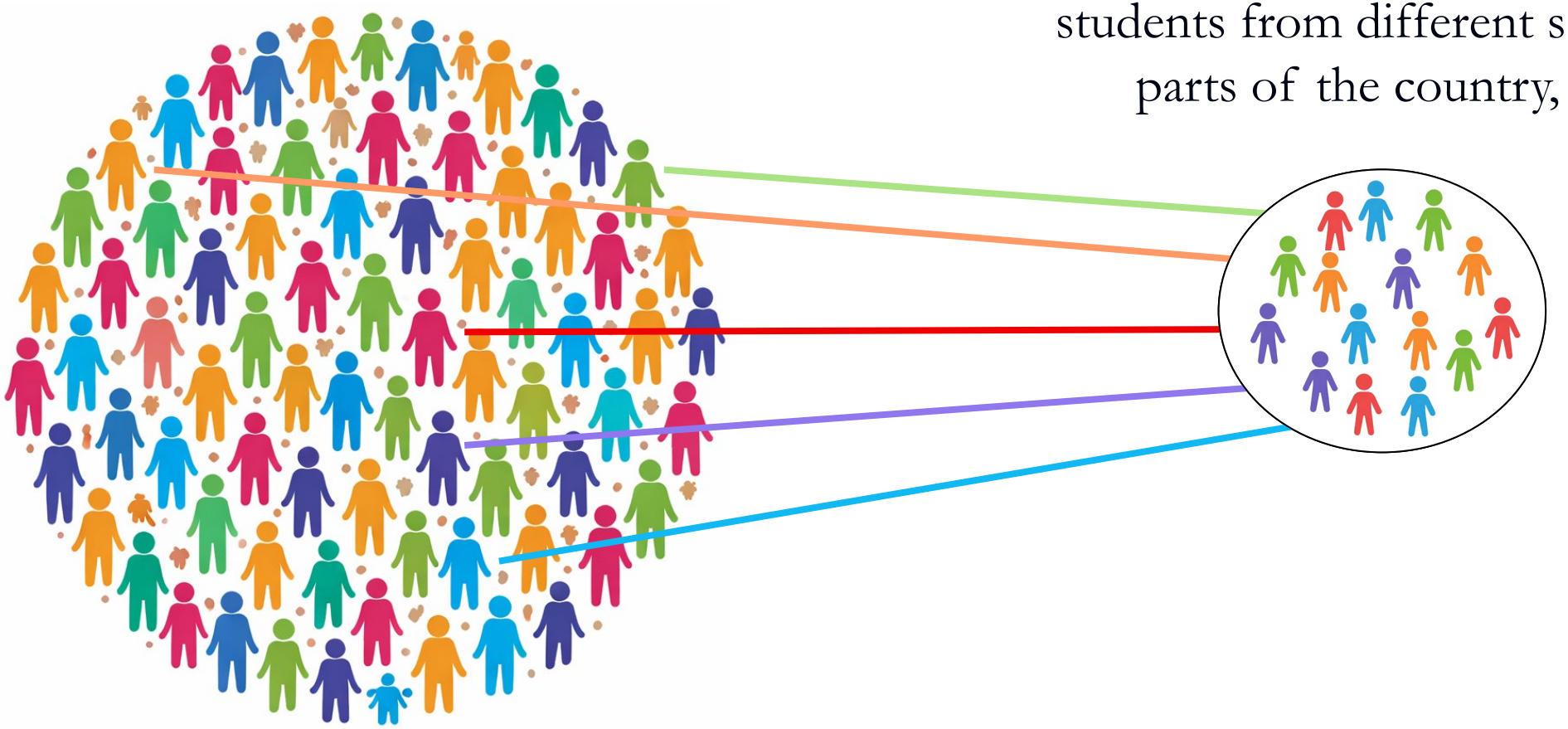
# Population vs. Sample
## Sample is a subset of the population

**Population**

All US business statistics students

**Sample**

Subset of business statistics students from different schools, parts of the country, etc.



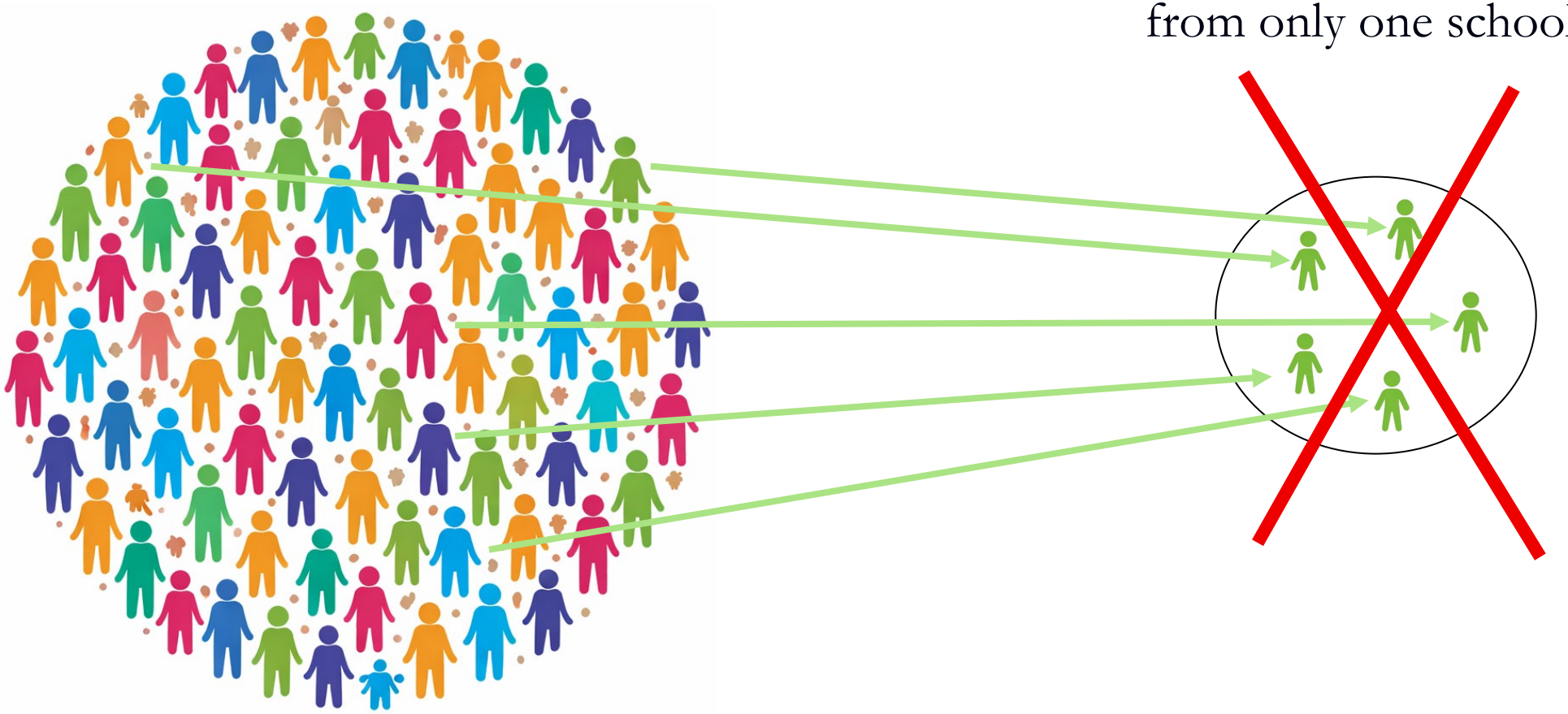
# Population vs. Sample
# All samples are not valid

**Population**

All US business statistics students

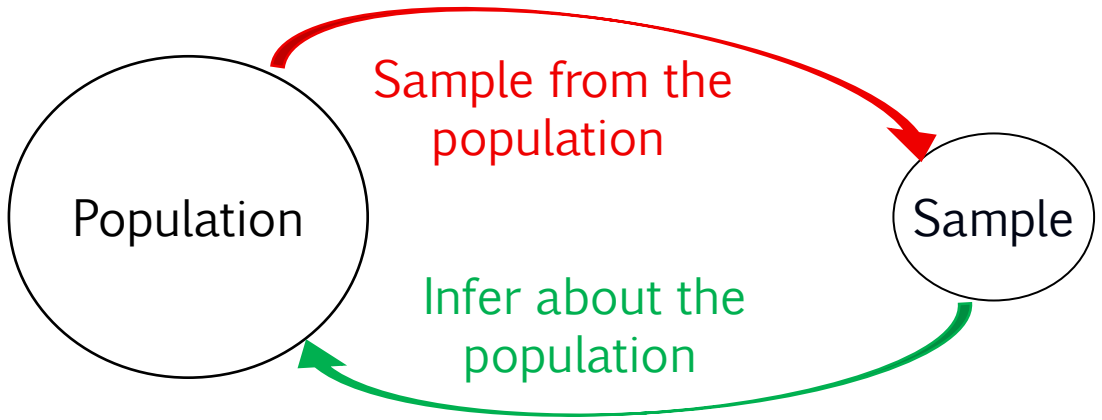
**Sample**

Business statistics students from only one school



# Population vs. Sample
# Descriptive and inferential statistics

- **Descriptive statistics** are about summarizing the data
- When all the data cannot be summarized, we must take a sample




- **Inferential statistics** are about using a sample to draw valid and meaningful conclusions about the population

# Population vs. Sample
# Measures: parameters vs. statistics

- Descriptive statistics are about “describing” the data
- The data will be described with measures like the average
- If describing a population, then those measures will be called **population parameters**
- If describing a sample, then those measures will be called **sample statistics**